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Influence of soil physicochemical properties on plant-parasitic nematode assemblages in contrasting agricultural soils

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ABSTRACT

This study investigates the spatial variation of plant-parasitic nematode (PPN) communities and their association with soil physicochemical properties across contrasting agricultural soils in three major regions of New Zealand: Waikato, Canterbury, and Manawatū-Whanganui. PPN assemblages were dominated by root-lesion and spiral nematodes, with Canterbury showing the highest Shannon and Simpson diversity and species richness, followed by Waikato and Manawatū-Whanganui. Non-metric multidimensional scaling (NMDS) and hierarchical clustering revealed partial regional separation of nematode communities, and multivariate analyses confirmed significant regional variation ($p < 0.05$). Spearman correlations indicated weak to moderate associations between PPNs, diversity, and soil properties: lesion nematodes were negatively correlated with pH and moisture, root-knot nematodes were weakly positively correlated with Shannon diversity and richness, and *Tylenchus* showed weak positive associations with diversity and C/N ratio. Diversity increased with silt but decreased with sand, phosphorus, carbon, and nitrogen. Additionally, canonical correspondence analysis (CCA) demonstrated that soil parameters collectively explained 20–25% of the variation in PPN community structure, with significant influences from phosphorus, pH, sand, clay, C/N ratio, and moisture content. Key associations, as indicated by their proximity to the corresponding environmental vectors, were observed between lesion nematodes and clay, spiral nematodes and moisture content, and ring nematodes and soil moisture and sand%. These findings provide the first detailed account of PPN ecology in New Zealand, highlighting the significant influence of regional soil characteristics on the composition of PPN communities. While valuable baseline data are presented, further research is recommended to explore seasonal dynamics, long-term trends, and the impact of different agronomic practices, informing more effective, region-specific PPN management strategies.

1. Introduction

A recent survey of New Zealand maize fields revealed that the prevalence, abundance, and diversity of plant-parasitic nematode (PPN) communities vary across fields. Additionally, root-lesion (*Pratylenchus* spp.) and spiral (*Helicotylenchus* spp.) nematodes are widely distributed and abundant compared with the other recovered genera. Some genera are confined geographically, such as ring nematodes (*Criconemella* spp.) in the Waikato region and root-knot (*Meloidogyne* spp.) and pin (*Paratylenchus* spp.) nematodes in Canterbury (Thiruchchelvan et al., 2024). These patterns are influenced by multiple factors, including climatic variation affecting PPN population dynamics and distribution (Efthimiadou et al., 2009), as well as soil characteristics, nematode virulence, host genotype, and farming practices (Alby et al., 1983; Chen et al., 2012; Edwards et al., 1988; Griffin, 1996; Kandel et al., 2013; Norton, 1983; Simon, 2015).

Studies have indicated that soil properties have a greater impact than all other ecological factors that influence the PPN communities (Neher, 1999; Yeates et al., 1999). Soil texture was the main driver of PPN reproduction and movements (Griffin, 1996; Koenning et al., 1996).

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Large-sized nematodes such as lance (*Hoplolaimus* spp.), and dagger (*Xiphinema* spp) nematodes were widely distributed and survived well in sandy soils, though other small-sized nematodes such as root-lesion and spiral nematodes favour clay soils (Wang and McSorley, 2005). Previous studies have shown that light, sandy soils have the highest populations of PPN compared to clay soils (Yeates, 1984; Yeates et al., 1997). Soil texture strongly influences the reproduction and infectivity of root-knot nematodes (*Meloidogyne* spp.), with coarse, sandy or sandy-loam soils characterized by high sand content and low clay/silt fractions being markedly more conducive to penetration, gall formation and root-knot nematode reproduction than heavy clay soils (Adegbite, 2017). According to Buckley and Schmidt (2003), plants that grew in well-drained soils are more likely to suffer from intermittent drought and are thus more vulnerable to damage by PPN. Desert valleys and tropical sandy soils are particularly challenged by nematode overpopulation (Efthimiadou et al., 2009). Optimal soil moisture, neither saturated nor excessively dry, supports root-knot nematode survival and infectivity by maintaining adequate water films for movement while preserving aerobic conditions (Karssen and Moens, 2022). Studies have also shown that soil chemical properties, such as pH and nutrient content, are significant contributors to changes in PPN population density and dissemination (Fleming et al., 2016; Kandji et al., 2003; Koenning et al., 1996; Prot and Van Gundy, 1981). As the pH of the soil increased from acidic to neutral, the abundance of PPN also increased. In the soil pH range of 6.5 to 7.5, the PPN abundance was 2.5 times more than in the pH range below 4.5 (Nisa et al., 2021). Soil physicochemical factors such as higher sand percentage, adequate soil moisture, and favourable nutrient status (e.g. phosphorus and potassium) have also been positively correlated with higher nematode population densities, whereas increased silt + clay fraction and lower moisture or unfavourable pH tend to suppress root-knot nematode development (Nishantha et al., 2020).

Further, soil nutrients such as N, P, and the C/N ratio influenced the PPN abundance in the field, where PPN increased with the increase in soil N and P content (Nisa et al., 2021). Macro-elements, such as calcium, and micro-elements, including iron, in addition to other climatic factors like rainfall, soil temperature, and soil water content, had an impact on the population densities of PPNs (Castillo et al., 1996; Govaerts et al., 2007; Karuri et al., 2017). Plant-parasitic nematode populations are typically more prevalent in warmer regions, where longer growing seasons extend their feeding periods and increase reproductive rates (Wang and McSorley, 2005). In colder regions, the life cycle of the nematodes tends to be prolonged by up to two weeks, compared to the life cycle length of 21 days. In maize cropping systems, the damage and reproduction potential of root-knot nematodes showed an increase during the rainy season, especially in seasons where an unnaturally high rainfall occurred (Mielie, 2014). *Pratylenchus zeae* has a shorter life cycle, faster reproductive rate, and higher tolerance to a wider range of temperatures than *P. brachyurus*. Moisture is also an important factor affecting the reproduction and damage potential of these root-lesion nematodes, which can increase during the rainy season (Mielie, 2013). Moreover, Nisa et al. (2021) reported 21–32 °C temperature and 35–40% soil moisture favourable for the PPN abundance, and a lower abundance of PPN was recorded at −1 °C temperature and soil moisture below 25%. As mentioned above recent survey in New Zealand maize fields indicated the prevalence and abundance of PPN significantly varied in each region, district, field and even within the field. Surveyed fields are represented across three regions: Canterbury in the South Island and the Waikato and Manawatu-Whanganui regions in the North Island of New Zealand (Thiruchchelvan et al., 2024). These regions are geographically and climatically different from each other (NIWA, 2023).

In New Zealand, there is limited information on the composition of PPNs and their relationship with soil characteristics, particularly in maize cultivation systems. Understanding how soil physicochemical properties influence nematode populations is crucial for effective PPN management. This study hypothesized that soil physicochemical properties affect the diversity and abundance of PPN genera, including Lesion (*Pratylenchus* spp.), spiral (*Helicotylenchus* spp.), root-knot (*Meloidogyne* spp.), cyst (*Heterodera* spp.), pin (*Paratylenchus* spp.), ring (*Criconemella* spp.) and *Tylenchus* spp. associated with maize fields. To test this hypothesis, the objectives of this study were to: (1) identify the PPN diversity, and (2) examine the relationships between PPN populations and key soil physicochemical properties across different regions of New Zealand.

2. Materials and methods

2.1. Sampling sites and soil collection

Between June and November 2022, a total of 384 composite soil samples were collected from 25 maize grain fields in three key regions of New Zealand: Canterbury (CaR), Waikato (WaR), and Manawatu-Whanganui (MWR), covering both islands. Depending on field size, samples were collected using either a double- or triple-zigzag pattern. In total, 10–25 composite samples per field were taken and processed separately. At each sampling point, 15 soil cores (30 cm deep, 2 cm diameter) were collected within a 4 m radius using an OAKFIELD soil sampler, yielding about 750 g of soil per composite sample.

2.2. Nematode extraction and identification

Nematodes were extracted using a modified Jenkins (1964) sieving and sugar-flotation method. From each sample, 100 g of soil was mixed with water, stirred, allowed to settle, and poured through 150 μm and 38 μm sieves. The residue on the 38 μm sieve was centrifuged at 576 ×*g* for 5 min, the pellet mixed with 45% sucrose, then centrifuged again for 1 min. The supernatant was passed through a 38 μm sieve, rinsed, and nematodes were collected into sterile tubes and stored at 4 °C Plant-parasitic nematodes were identified to genus level based on key morphological characters including head shape, stylet and stylet-knob morphology, oesophageal structure, intestinal overlap, female tail terminus and vulval position, following the identification criteria of Fortuner (1988) and Mai et al. (1996). Nematodes were heat-killed on glass slides and examined under a compound light microscope (Olympus CKX53, Japan) at ×40–100 (and up to ×600 when required) magnification. Only nematodes extracted from soil samples were counted; root samples were not processed.

2.3. Soil physicochemical analysis

A total of 384 soil samples were analysed between June 2022 and August 2023 at the Analytical Laboratory, Lincoln University, to determine key physicochemical properties. Soil moisture content was assessed using the gravimetric oven-drying method at 105 °C for 24 h, following the method described by Blakemore et al. (1987) and Rayment and Lyons (2010). For air-dried analyses, soil subsamples (~200 g) were dried at 25 °C, sieved through a 2 mm mesh, and used for subsequent tests. Soil pH was measured in a 1:2.5 (soil: water) suspension using a calibrated pH meter (Mettler Toledo Seven Easy S20) after shaking and overnight settling, as described by Blakemore et al. (1987). Soil texture was determined using the micropipette method (Miller and Miller, 1987), involving dispersion in sodium hexametaphosphate, timed settling, and oven drying of clay and sand fractions to calculate relative percentages of sand, silt, and clay. Nutrient analysis included the determination of available phosphorus (Olsen P) using 0.5 M $NaHCO_3$ extraction followed by spectrophotometric measurement (SMARTCHEM 200) according to Olsen et al. (1954) with modifications by Murphy and Riley (1962). Total carbon and nitrogen contents were analysed using an Elementar Vario-Max CN Elemental Analyser (Germany).

2.4. Statistical analysis

2.4.1. Nematode abundance data

Because not all nematode genera occurred in every region, a non-parametric Kruskal–Wallis test with Dunn's post hoc test was used to compare nematode populations among regions. These statistical analyses were conducted only for the major nematode groups with adequate regional representation: Non-PPN, PPN, lesion nematodes, and spiral nematodes. Nematode abundances by region for each genus were reported as mean ± standard error (SE).

2.4.2. Data processing and diversity metrics

Samples with zero PPN abundance across all genera were excluded from analyses. Diversity indices, including Shannon, Simpson, and species richness, were calculated for each sample using the vegan package in R (Oksanen et al., 2022; Development Core Team R, 2013).

2.4.3. Community composition analyses

The nematode community composition was analysed using non-metric multidimensional scaling (NMDS) based on Bray-Curtis dissimilarities (Bray and Curtis, 1957). NMDS was performed with two dimensions and 300 iterations, applying square-root transformation and Wisconsin double standardization to stabilize variance (Clarke, 1993). Differences in community composition among regions were evaluated using permutational multivariate analysis of variance (PERMANOVA; adonis2 function) and analysis of similarities (ANOSIM) with 999 permutations (Anderson, 2017; Clarke, 1993).

2.4.4. Correlation analyses

Spearman rank correlations were calculated between nematode abundances, diversity indices, and soil physicochemical properties. Soil properties included moisture content (MC%), pH, clay, sand, silt, phosphorus (P), nitrogen (N), carbon (C), and C/N ratio. Correlation matrices were visualized using *corrplot* to identify potential relationships between nematodes and soil characteristics (Wei and Simko, 2021).

2.4.5. The similarity in PPN communities in New Zealand maize fields

The similarity of PPN communities in New Zealand maize fields was compared using the Bray-Curtis similarity index (Bray and Curtis, 1957; Chen et al., 2012; Chowdhury et al., 2020). Among the regions (Canterbury, Waikato, and Manawatu-Whanganui), the districts such as Ashburton, Manawatu, Matamata-Piako, Otorohanga, Rangitikei, Selwyn, Waikato, Waimakariri, and Waipa districts and finally among the 24 maize fields that were sampled in the 2022 survey. The Bray-Curtis similarity index was estimated using the package "Vegan" in R statistical software (Development Core Team R, 2013; Karuri et al., 2017; Oksanen et al., 2022). The dissimilarity (using the Bray-Curtis dissimilarity index (1-similarity) between regions, districts, and fields was visualized using the dendrograms, followed by the hierarchical cluster analysis (complete linkage) using the 'hclust' function of the Vegan package in R (Bray and Curtis, 1957; Development Core Team R, 2013; Oksanen et al., 2022).

2.4.6. Soil properties and PPN community abundance association

Canonical correspondence analysis (CCA) was used to examine the relationships between biological data (PPN composition) and six standardized environmental variables (P, pH, moisture content, clay, sand, and C/N ratio) in New Zealand maize fields and within each region (Chowdhury et al., 2020; Karuri et al., 2017; Ter Braak and Verdonschot, 1995). Nematode population data were log (x + 1) transformed to reduce the heterogeneity of the data, as suggested (Chowdhury et al., 2020; Ter Braak and Verdonschot, 1995). Soil properties data presented as percentages and proportions (moisture content, N, C, clay, sand, and silt percentages and C/N ratio) were arcsine transformed; P and pH were log-transformed to achieve the normality of the environmental data as suggested (Ter Braak and Verdonschot, 1995). To avoid redundancy and improve interpretability, multicollinearity among soil variables was assessed, and highly collinear variables (C, N, and silt) were excluded from the final CCA. Analysis was performed using the "Vegan" package in R software (Chowdhury et al., 2020; Development Core Team R, 2013; Oksanen et al., 2022; Ter Braak and Verdonschot, 1995) to plot the data into a CCA plot (ordination plot) representing the three sets of variables in a plot, such as environmental, PPN community, and the respective location of the sample. Monte Carlo permutation test (unrestricted permutation 999) was used to evaluate the significance ($p < 0.05$) of the environmental variables in PPN communities.

3. Results

3.1. Abundance of PPN in New Zealand

Non-PPN and PPN abundances differed significantly across the three regions (Fig. 1a, b; $\chi^2(2) = 63.05, p < 0.001$ for non-PPN; $\chi^2(2) = 59.52$, $p < 0.001$ for PPN). Across regions, non-PPN were the dominant group, with the highest mean abundance recorded in Canterbury (9744 ± 536 kg^{-1} soil), followed by Waikato (5641 ± 284 kg^{-1}) and Manawatū-Whanganui (5452 ± 390 kg^{-1}). A similar pattern was observed for total PPN, which also peaked in Canterbury (2273 ± 180 kg^{-1}), relative to Waikato (1339 ± 138 kg^{-1}) and Manawatū-Whanganui (736 ± 127 kg^{-1}). Lesion and spiral nematodes also showed significant regional differences (Fig. 1c, d; $\chi^2(2) = 31.71$, $p < 0.001$ for lesion; $\chi^2(2) = 6.10$, $p = 0.047$ for spiral), with Canterbury exhibiting substantially higher abundances of both groups. Mean lesion nematode counts were highest in Canterbury (1158 ± 125), compared with Waikato (598 ± 56) and Manawatū-Whanganui (560 ± 83), while spiral nematodes were likewise most abundant in Canterbury (869 ± 139), moderate in Waikato (584 ± 134), and lowest in Manawatū-Whanganui (164 ± 59). Root-knot nematodes were detected only in Canterbury (174 ± 34), with negligible abundance in the other regions. Cyst nematodes (45 ± 11), pin nematodes (13 ± 6), and Tylenchus (15 ± 9) also occurred predominantly in Canterbury, whereas *Tylenchus* and ring nematodes were in Waikato (151 ± 37 and 5.7 ± 4.6, respectively) (Fig. 1).

3.2. Plant-parasitic nematode community composition

Non-metric multidimensional scaling (NMDS) revealed clear compositional separation of nematode communities among regions (stress = 0.14; Fig. 2). Samples from Canterbury occupied a distinct ordination space compared with Waikato and Manawatū–Whanganui, with minimal overlap (Fig. 2). PERMANOVA confirmed a significant regional effect on nematode community composition ($F = 13.53$, $R^2 = 0.068$, $p < 0.001$), also supported by ANOSIM ($R = 0.076$, $p < 0.001$).

Diversity indices varied by region (Table 1). Mean Shannon diversity ranged from 0.21 in Manawatū–Whanganui to 0.36 in Canterbury, whereas Simpson diversity ranged from 0.13 to 0.22 accordingly. Species richness was also highest in Canterbury (1.87 ± 0.27), intermediate in Waikato (1.66 ± 0.27), and lowest in Manawatū–Whanganui (1.51 ± 0.30). Additional analyses of dissimilarity of the PPN genera in New Zealand fields were supported by hierarchical cluster analyses based on Bray-Curtis dissimilarity at regional, district, and field levels, which provide further details on community similarities, and are presented as supplementary figures (Supplementary Figs. S1–S3). At the regional level, PPN community similarities ranged from 48% to 71%, with the Manawatu-Whanganui and Waikato regions being the most similar compared to Canterbury. At the district level, certain districts exhibited higher similarities, such as 92% between Manawatu and Waimakariri. Field-level dendrograms highlighted pronounced heterogeneity, with five subclusters observed at >50% similarity and one highly unique field, Tahuna, where only the spiral nematode was detected. Full Bray-Curtis similarity matrices are provided in Supplementary Tables 1 and 2.

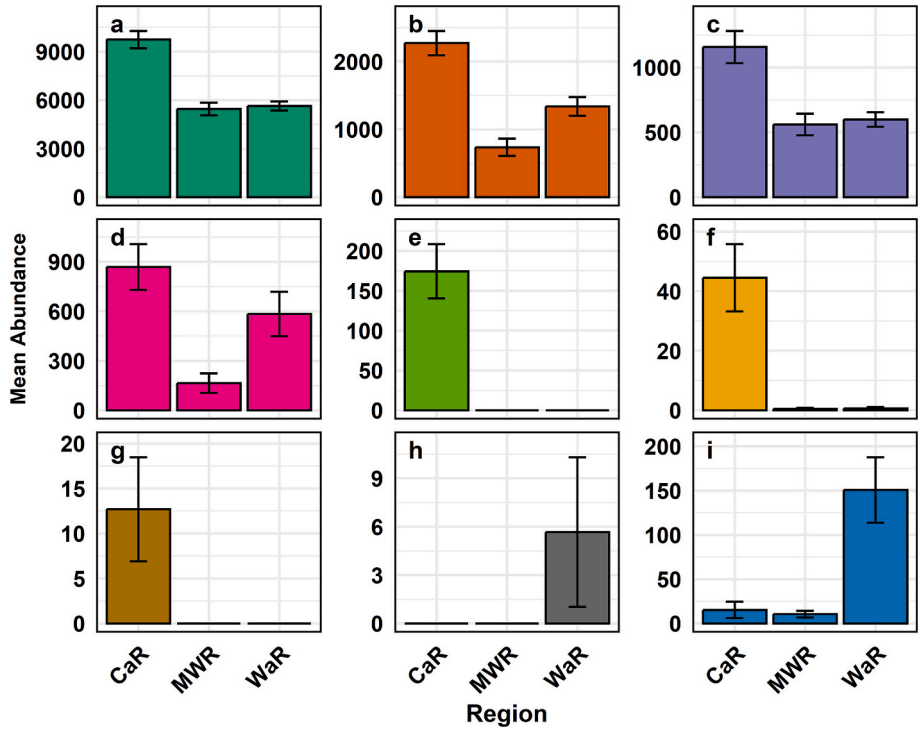

Fig. 1. Mean abundances of nematodes in Canterbury (CaR), Manawatu-Whanganui (MWR), and Waikato (WaR): (a) non-plant-parasitic nematodes, (b) plant-parasitic nematodes, (c) root-lesion nematodes, (d) spiral nematodes, (e) root-knot nematodes, (f) cyst nematodes, (g) pin nematodes, (h) ring nematodes, and (i) *Tylenchus* nematodes. Error bars represent the standard error of the mean.

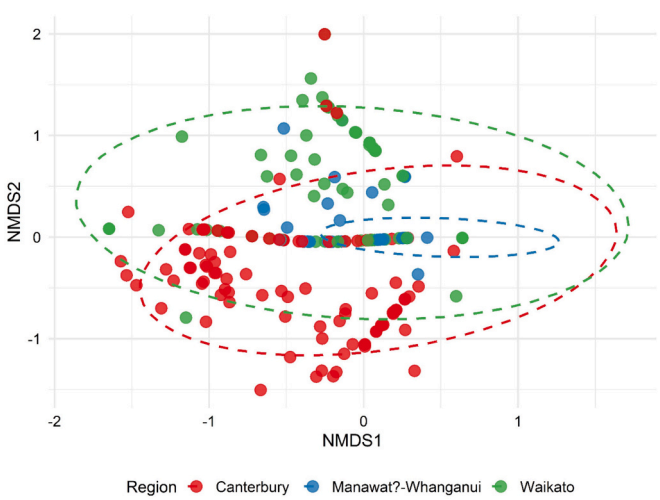

Fig. 2. Non-metric multidimensional scaling (NMDS) plot showing the composition of nematode communities across three regions: Canterbury (red), Manawatu-Whanganui (blue), and Waikato (green). Ellipses represent 95% confidence intervals for each region.

Table 1
Diversity indices between the regions of the New Zealand maize fields.

Region	Shannon (H′) ± SE	Simpson (D) ± SE	Richness (S) ± SE
Canterbury	0.355 ± 0.026	0.218 ± 0.016	1.865 ± 0.061
Manawatu–Whanganui	0.207 ± 0.033	0.134 ± 0.022	1.506 ± 0.075
Waikato	0.294 ± 0.029	0.187 ± 0.019	1.659 ± 0.060

3.3. Relationship between soil properties and PPN community

Spearman correlation analysis revealed a range of weak to moderate (0.2–0.5) associations between PPNs, diversity indices, and soil physicochemical properties (Fig. 3). Lesion nematodes were negatively correlated with soil pH ($\rho = -0.367$, $R^2 = 0.135$) and moisture content ($\rho = -0.240$, $R^2 = 0.057$). Other nematode-soil relationships were generally weak to moderate ($R^2 = 0.011$–0.057). For instance, root-knot nematodes exhibited weak positive correlations with Shannon diversity ($\rho = 0.423$, $R^2 = 0.179$) and species richness ($\rho = 0.422$, $R^2 = 0.178$). Similarly, Tylenchus showed weak positive associations with diversity indices and soil parameters, C/N ratio ($\rho = 0.363$, $R^2 = 0.132$). Diversity indices were significantly influenced by soil texture and fertility: Shannon and Simpson diversity were positively associated with silt content and negatively with sand, phosphorus (P), carbon (C), and nitrogen (N), while species richness was positively associated with silt content and weakly negatively associated with sand ($R^2 \leq 0.049$). Comparison of soil characteristics between regions of New Zealand in the supplementary Table 3.

3.4. Soil properties and PPN community abundance association

3.4.1. New Zealand maize fields (All 24 fields together)

Canonical correspondence analysis for New Zealand maize fields showed that 20% of the total inter-site variation in distribution and PPN population abundance was explained by the studied soil parameters (Fig. 4). A 51.22% (eigenvalue 0.20124) of the total variation in the PPN and soil characteristics relationship was explained by the canonical correspondence analysis axis 1 (CCA1), and 32.60% variation was explained by CCA2 (eigenvalue 0.12864). The permutation tests confirmed that the CCA model was highly significant ($F = 15.291$, $p = 0.001$). Also, both the CCA1 ($F = 46.997$, $p = 0.001$) and CCA2 ($F = 29.908$, $p = 0.001$) axes were statistically significant (and CCA3 $F = 9.116$, $p = 0.014$)). Permutation tests for individual soil variables showed that phosphorus (P; $F = 8.822$, $p = 0.001$), pH ($F = 5.455$, $p =$

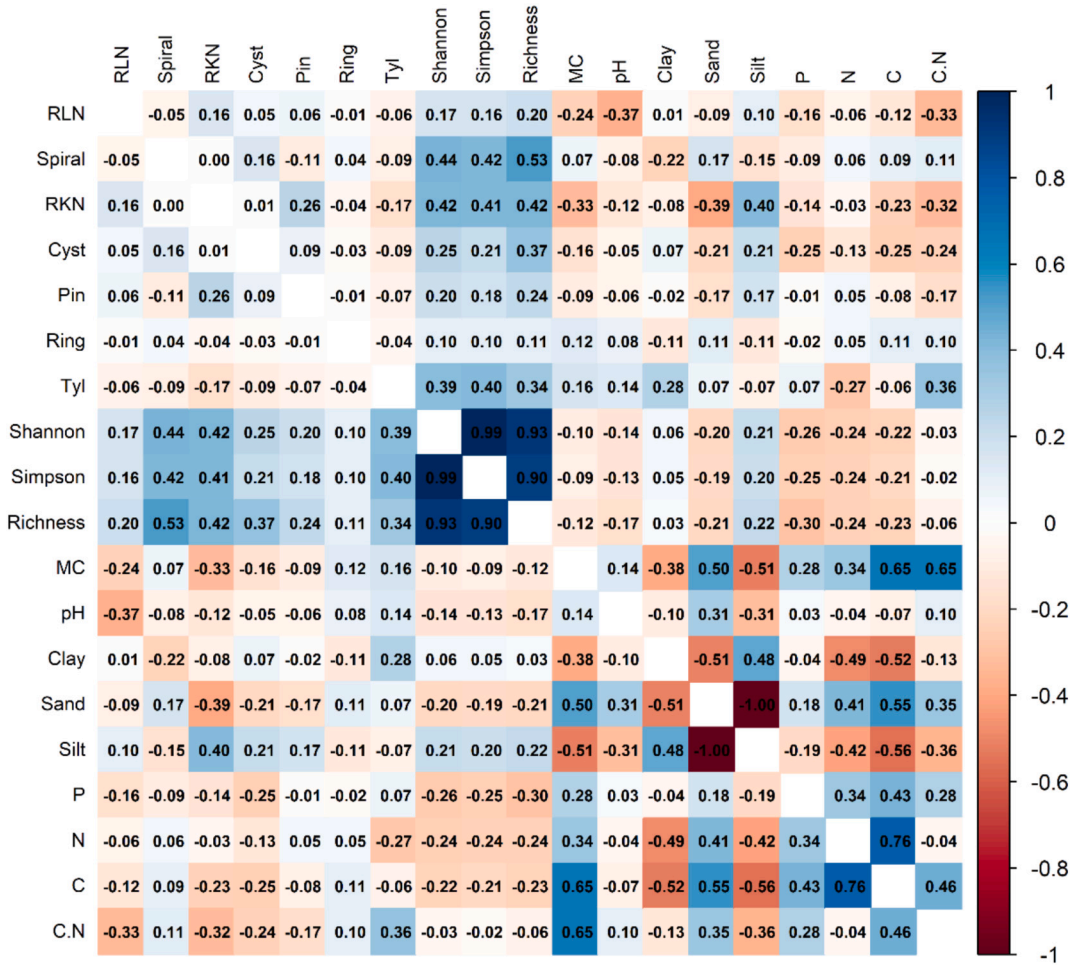

Fig. 3. Correlation matrix showing relationships among nematode groups([RLN]- root lesion, [RKN] root knot nematode [Tyl]– *Tylenchus* diversity indices (Shannon, Simpson, Richness), and soil physicochemical properties (moisture content [MC], pH, clay, sand, silt, phosphorus [P], nitrogen [N], carbon [C], and C/N ratio). Positive correlations are shown in blue and negative correlations in red, with intensity proportional to the correlation coefficient.

0.004), soil moisture content (MC; F = 25.752, $p = 0.001$), clay (%) (F = 17.692, p = 0.001), sand (%) (F = 20.085, $p = 0.001$), and C/N ratio (F = 13.941, $p = 0.002$) all had significant effects on PPN community structure. The CCA species scores indicated that lesion nematodes (*Pratylenchus* spp.) were negatively correlated with both CCA1 and CCA2, whereas spiral nematodes (*Helicotylenchus* spp.) were positively correlated with both axes. Root-knot (*Meloidogyne* spp.), cyst (*Heterodera* spp.), and pin (*Paratylenchus* spp.) nematodes were positively correlated with CCA1 and negatively correlated with CCA2. Ring nematodes (*Criconemella* spp.) were negatively correlated with CCA1 and positively correlated with CCA2, while *Tylenchus* spp. were negatively correlated with both CCA1 and CCA2.The biplot of the soil characteristics (Fig. 4, with green arrows), whereby the length of the arrow indicates the strength of the correlation of the characteristic with PPN genera composition. According to the biplot scores, P, pH, sand%, moisture%, clay%, and C/N ratio, negatively correlated with CCA1. All the soil characteristics positively correlated with the CCA2, except the clay% (negative correlation with CCA2). The biplot indicates the proximity of the soil characteristics to the PPN genera. The soil moisture content and sand% on ring nematode, clay% on the lesion nematode and soil moisture on spiral nematode abundances had a significant association.

3.4.2. *Relationship between soil properties and PPN community in Canterbury*

Canonical correspondence analysis results of 6 soil characteristics influencing five PPN genera populations at maize fields in the Canterbury region are illustrated in Fig. 5. There was 24% inter-site variation of the PPN genera that was explained by 6 soil variables, and the CCA model was highly significant (F = 8.886, $p = 0.001$). The total variation of the relationship between the PPN genera and soil characteristics was explained by the axis CCA1, which was 53.1% (eigenvalue = 0.1450; F = 28.634, p = 0.001), and CCA2 was 32.3% (eigenvalue = 0.08841; F = 17.452, p = 0.002). Permutation tests for individual soil variables showed that P (F = 13.175, p = 0.001), pH (F = 5.913, p = 0.002), MC (F = 4.924, $p = 0.011$), clay (%) (F = 10.761, p = 0.001), sand (%) (F = 12.885, p = 0.001), and C/N ratio (F = 5.657, $p = 0.004$) all had significant effects on the PPN community structure. Scores of the CCA for the PPN genera revealed that the spiral, root-knot, cyst, and pin nematodes were positively correlated with CCA1. Lesion nematode negatively correlated with CCA1 and positively correlated with CCA2. Also, root-knot and pin nematodes positively correlated with CCA2, and spiral and cyst nematodes were negatively correlated with CCA2.

3.4.3. *Relationship between soil properties and PPN community in Waikato*

The influence of soil characteristics on PPN genera abundance in the Waikato region maize fields was demonstrated using the CCA ordination plot in Fig. 6. The analysis revealed that 24.37% of the total inter-site variation was explained by the relationship between soil characteristics and the abundance of PPN genera. The overall CCA model was statistically significant (F = 6.45, p = 0.001), indicating a meaningful relationship between soil characteristics and nematode assemblages.

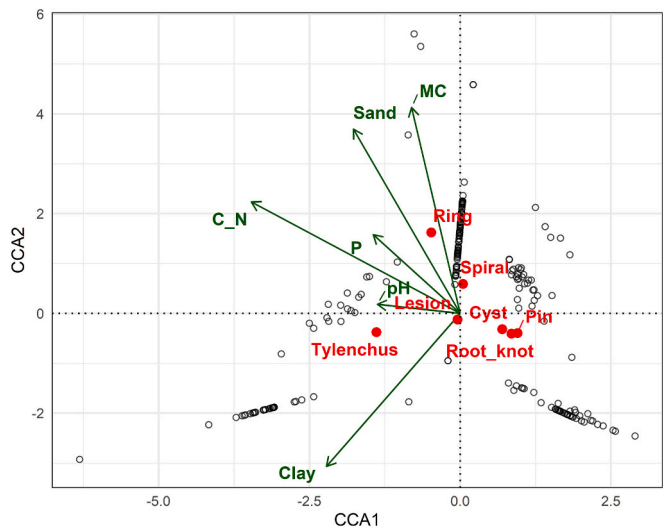

Fig. 4. Relationships between soil characteristics and PPN genera in New Zealand maize fields (CCA ordination plot).
*Sample sites are indicated by the black circles. Nematode genera are indicated by the red dots with common names, which include Spiral-(Helicotylenchus), Pin (Paratylenchus), Lesion (*Pratylenchus*), Tylenchus (*Tylenchus*), Cyst (*Heterodera*), Ring (*Criconemella*) and Root-knot (*Meloidogyne*). The green arrows represent the environmental variables (sand%, clay%, Moisture content (MC), pH, available P (P), and C_N - C/N ratio).*

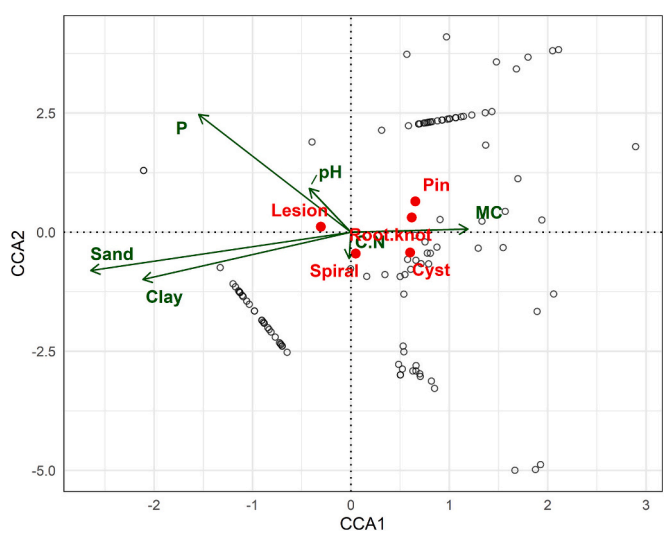

Fig. 5. Relationships between soil characteristics and PPN genera in maize fields of the Canterbury region (CCA ordination plot).
*Sample sites are indicated by the black circles. Nematode genera are indicated by the red dots with common names, which include Spiral-(*Helicotylenchus*), Pin (*Paratylenchus*), Lesion (*Pratylenchus*), Cyst (*Heterodera*), and Root-knot (*Meloidogyne*). The green arrows represent the environmental variables (sand%, clay%, Moisture content (MC), pH, available P (P), C·N - C/N ratio.*

The CCA1 accounted for 78.36% of the explained constrained variation (eigenvalue = 0.276), while the second axis (CCA2) explained 20.78% (eigenvalue = 0.073). Permutation tests confirmed that CCA1 was highly significant (F = 30.81, p = 0.001), whereas CCA2 was not statistically significant (F = 8.169, p = 0.070). Soil characteristics such as P (F = 8.81, p = 0.001), MC (F = 15.15, p = 0.001), and clay (F = 7.39, p = 0.008) exerted significant influences on PPN community structure. In contrast, soil pH (F = 2.3645, p = 0.078), sand (F = 2.5639, p = 0.108), and C/N ratio (F = 2.3904, p = 0.078) did not show statistically significant effects. (See Fig. 6.)

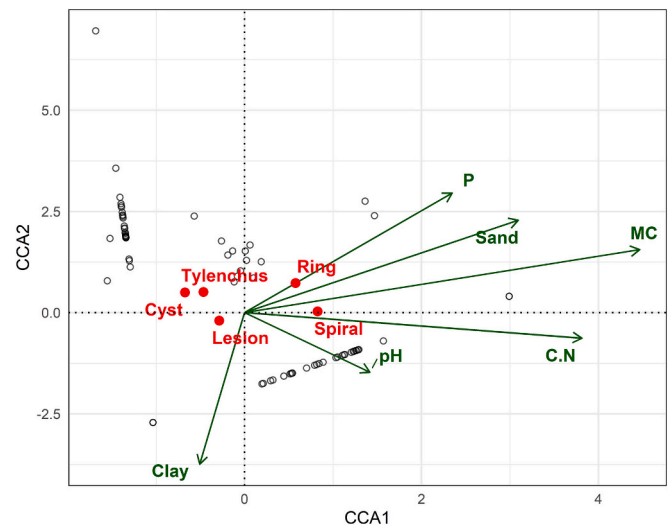

Fig. 6. Relationships between soil characteristics and PPN genera in maize fields of Waikato region (CCA ordination plot).
*Sample sites are indicated by the black circles. Nematode genera are indicated by the red dots with common names, which include Spiral-(*Helicotylenchus*), Lesion (*Pratylenchus*), Cyst (*Heterodera*), Tylenchus (*Tylenchus*) and Ring (*Cricinemella*). The green arrows represent the environmental variables (sand%, silt%, clay %, Moisture content (MC), pH, available P (P), total C, N, –and C·N - C/N ratio.*

3.4.4. Relationship between soil properties and PPN community in the Manawatu-Whanganui region

The CCA showed that soil physicochemical variables explained 13.5% of the total variation in the biological data (constrained inertia = 0.1196 of total inertia = 0.8862, Fig. 7). However, the overall CCA model was not statistically significant based on permutation testing (F = 1.92, p = 0.095). The first two canonical axes explained most of the constrained variation, with CCA1 accounting for 66.8% and CCA2 for 31.3% of the explained variance. Axis-wise permutation tests indicated that CCA1 (F = 8.02, p = 0.087), and CCA2 (F = 3.76, p = 0.764) were not significant. Among the individual soil variables, moisture content

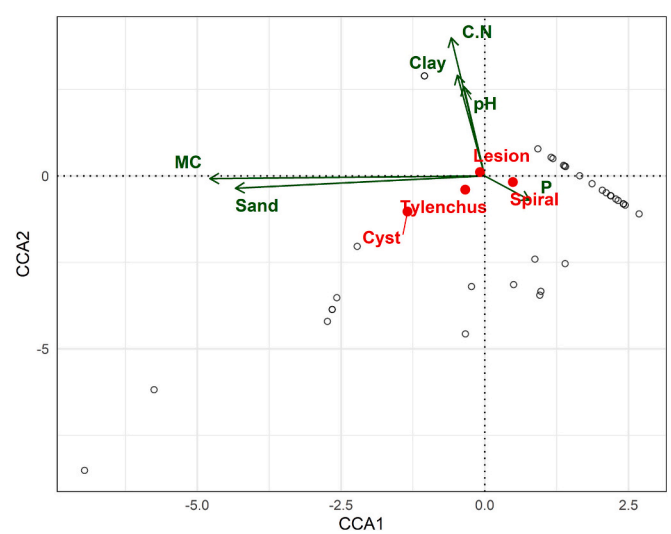

Fig. 7. CCA ordination plot for relationships between soil characteristics and PPN genera in maize fields of the Manawatu-Whanganui region.
*Sample sites are indicated by the black circles. Nematode genera are indicated by the red dots with common names, which include Spiral-(*Helicotylenchus*), Lesion (*Pratylenchus*), Cyst (*Heterodera*), and Tylenchus (*Tylenchus*). The green arrows represent the environmental variables (sand%, clay%, Moisture content (MC), pH, available P (P), and C·N - C/N ratio.*

was the only factor that showed a significant effect on the biological data ($F = 5.33$, $p = 0.007$). All other soil variables, including phosphorus, pH, clay, sand, and C/N ratio, did not have significant effects ($p > 0.05$).

4. Discussion

This is the first New Zealand–wide baseline characterisation of PPN abundance, diversity, and community structure in maize fields across contrasting agroecological regions. By integrating community profiling, multivariate ordination and soil physicochemical analyses, the results demonstrate distinct geographic signatures in PPN assemblages and provide new evidence linking nematode distribution to soil texture, fertility, and moisture conditions. These findings fill a major knowledge gap for New Zealand maize production systems, where regional distribution patterns and their environmental drivers have not been previously evaluated.

Marked regional variation in PPN abundance and richness, with Canterbury consistently showing higher counts of lesion and spiral nematodes, as well as the detection of root-knot nematodes. This pattern suggests that Canterbury represents a hotspot of PPN pressure in New Zealand maize systems, most likely influenced by specific combinations of soil texture, climate and cropping history. Comparable regional differentiation in PPN assemblages has been reported elsewhere and attributed to climatic constraints, soil structure, and long-term agronomic practices that shape nematode niches (Neher, 2001; Yeates and Bongers, 1999; Ferris, 2010). The exclusive presence of root-knot nematodes in Canterbury may reflect the compatibility of sandy soils, free drainage, together with persistent host availability arising from continuous maize cultivation and the presence of alternative host plants (e.g., rotational crops) that allow *Meloidogyne* populations to survive between growing seasons(Karssen and Moens, 2022), and highlights a potential biosecurity concern for the wider maize industry.

Diversity indices also differed between regions, with higher Shannon and Simpson diversity values in Canterbury and lower diversity in Manawatū–Whanganui. These differences suggest broader niche partitioning and potentially more stable or complex soil food webs under the environmental conditions of the South Island. Previous studies have similarly shown that soil fertility, climatic regime and disturbance history strongly affect nematode diversity and trophic organisation (Neher, 2001; Bongers and Ferris, 1999). Higher PPN diversity may reflect a more diverse host base created by crop rotations, as well as greater soil heterogeneity in Canterbury cropping landscapes.

Associations between soil physicochemical properties and nematode communities were generally weak to moderate, yet they reveal ecologically meaningful patterns. Lesion nematodes were negatively related to soil pH and moisture, consistent with earlier observations in temperate cropping systems (McSorley, 2011), while spiral nematodes showed positive associations with nitrogen content, which may reflect their tendency to proliferate under elevated fertility (Yeates and Bongers, 1999). Root-knot nematodes exhibited positive correlations with community diversity and richness, likely due to their broad host range and high reproductive capacity, which allow them to persist and reproduce across multiple host plants present in maize-based crop rotations and associated weeds, particularly under favourable soil physical conditions such as well-drained, sandy soils that facilitate juvenile movement and root infection (Karssen and Moens, 2022). These relationships indicate that although soil parameters do not entirely control PPN composition, they contribute important constraints consistent with the multicausal drivers shaping PPN biogeography (Ferris, 2010).

Diversity was positively associated with silt and negatively with sand, implying that finer-textured soils support more complex nematode assemblages. Soil texture influences moisture, aeration, root growth and microbial processes, thereby regulating soil food-web structure (Bongers, 1990). The influence of carbon, nitrogen and silt further suggests that organic matter and soil structure jointly shape PPN organisation in New Zealand maize systems. Hierarchical clustering revealed considerable spatial heterogeneity, with higher similarity between Waikato and Manawatū–Whanganui and much greater dissimilarity with Canterbury, reflecting climatic and edaphic contrasts (Chowdhury et al., 2020; Karuri et al., 2017). The highly distinct Tahuna field, where spiral nematode dominated, underscores the importance of fine-scale soil variation, congruent with the strong microsite effects commonly reported for nematode communities (Simon, 2015; Simon et al., 2018).

The CCA results demonstrate pronounced regional variability in soil–PPN relationships across New Zealand maize fields. When all sites were analysed together, soil physicochemical properties explained a significant proportion of variation in PPN community structure, indicating that broad gradients in soil texture, moisture, and nutrient status exert strong controls on nematode assemblages at a national scale (Koenning et al., 1996; Neher, 2001). Likewise, in the Canterbury and Waikato regions, significant CCA models and dominant first axes suggest that soil texture and moisture-related variables are key drivers of PPN distribution, consistent with their known effects on nematode movement, survival, and host–root interactions (McSorley, 2011). In contrast, the Manawatu–Whanganui region showed a weak and non-significant overall soil–PPN relationship, with only soil moisture content exhibiting a significant effect, suggesting that PPN community structure in this region may be governed by additional factors, such as crop management practices, host availability, or historical land use (Chowdhury et al., 2020; Neher, 2001). Collectively, these findings highlight that while soil characteristics strongly influence PPN communities at broader spatial scales, their relative importance is region-specific, underscoring the need for localized assessments of nematode–soil interactions in maize agroecosystems.

The finding that soil characteristics influencing nematode assemblages differed across regions underlines that PPN ecology is context dependent and shaped by interactive effects of soil chemistry, texture, fertility, host availability and climate (Norton and Hoffmann, 1974; Neher, 1999; Duarte et al., 2024). The consistently higher PPN load in Canterbury, including the presence of root-knot nematodes, highlights the need for regionalised nematode management. Lesion, spiral and root-knot nematodes are known to predispose maize roots to secondary pathogens and reduce nutrient uptake, ultimately affecting plant growth (Nicol et al., 2011). Regionally tailored strategies, including crop rotation, soil amendments, organic additions and practices that enhance suppressive microbial communities, could help mitigate nematode pressure (Stirling, 2014; McSorley, 2011). Moreover, the associations between soil properties and PPN abundance suggest that management strategies targeting soil texture modification, organic matter inputs, or balanced fertilisation may weaken nematode establishment or competitiveness.

Although this study provides essential baseline information, it represents a single sampling period and therefore does not account for seasonal variability in nematode dynamics. Given that temperature and moisture strongly influence nematode behaviour (Duarte et al., 2024), future research should incorporate seasonal sampling and detailed management histories. Integrating microbial community profiling and food-web indices would further clarify functional relationships among soil biota, nutrient cycling, and PPN assemblages (Ferris et al., 2012). Nevertheless, the results presented here establish a foundation for understanding PPN risks in New Zealand maize systems and emphasise the role of region-specific soil and environmental conditions in shaping nematode community structure.

5. Conclusions

This study is the first region-wide assessment of PPN communities in New Zealand's maize systems, demonstrating marked regional differences driven largely by variations in soil physicochemical properties. Canterbury emerged as a hotspot of PPN abundance and diversity, including the exclusive presence of root-knot nematodes, highlighting substantial regional biosecurity and management implications. While

soil properties exerted modest but detectable influences on community structure, other agronomic and climatic factors likely contribute to the observed patterns. These findings provide a necessary baseline for developing region-specific nematode management strategies and indicate the need for continued monitoring and systematic research.

CRediT authorship contribution statement

Nagarathnam Thiruchchelvan: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Manjula Kularathna:** Writing – review & editing, Validation, Supervision, Software, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Romy Moukarzel:** Writing – review & editing, Validation, Supervision, Software, Resources, Formal analysis, Conceptualization. **Leo Condron:** Writing – review & editing, Validation, Supervision, Software, Resources, Project administration, Formal analysis, Data curation, Conceptualization. **Seona Casonato:** Writing – review & editing, Validation, Supervision, Software, Resources, Project administration, Data curation, Conceptualization.

Ethical approval

Not applicable.

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Declaration of competing interest

The authors declare the following financial interests/personal relationships, which may be considered as potential competing interests:

Nagarathnam Thiruchchelvan reports that financial support was provided by the World Bank. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.geodrs.2026.e01056.

Data availability

All related data are included in this article.

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